```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")
```

```
df = pd.read_csv("train.csv")
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs	female	38.0	1	0	PC 17599

Next steps: Generate code with df New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

Observation:

The dataset contains both numerical and categorical variables.

Columns like Age, Cabin, and Embarked have missing values.

Survival is a binary target variable.

```
df.describe()
```

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	89
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	3
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	4
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	1
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	3
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	51

Observation:

Average passenger age is around 30 years.

Fare values vary widely, indicating skewness.

Survival rate is below 50%, showing class imbalance.

```
df['Survived'].value_counts()
df['Sex'].value_counts()
df['Pclass'].value_counts()
```

Pclass	count
3	491
1	216
2	184

dtype: int64

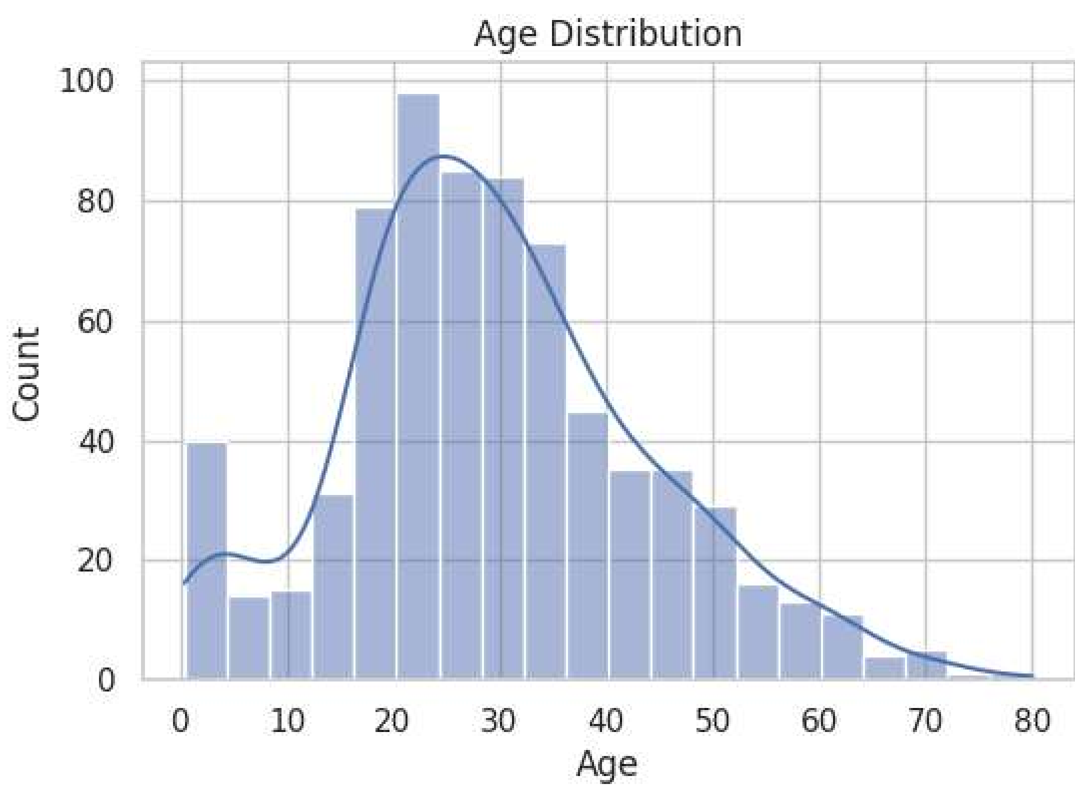
Observation:

More passengers did not survive than survived.

Male passengers outnumber females.

Most passengers traveled in third class.

```
plt.figure(figsize=(6,4))
sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()
```



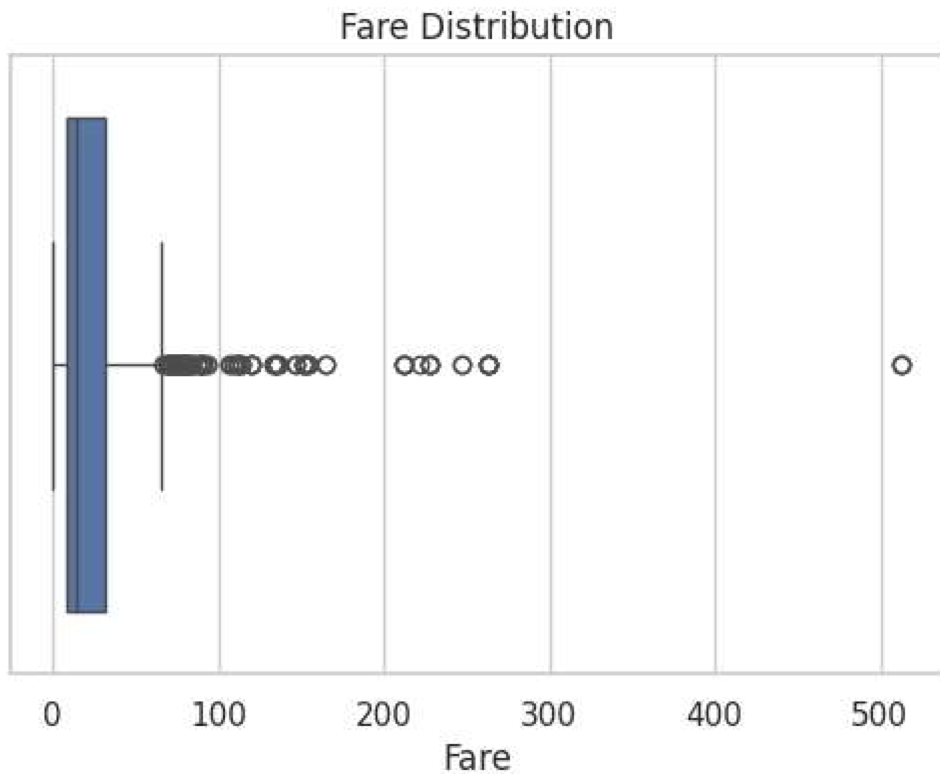
Observation:

Most passengers were between 20 and 40 years old.

The distribution is slightly right-skewed.

Children and elderly passengers were also present.

```
plt.figure(figsize=(6,4))
sns.boxplot(x=df['Fare'])
plt.title("Fare Distribution")
plt.show()
```

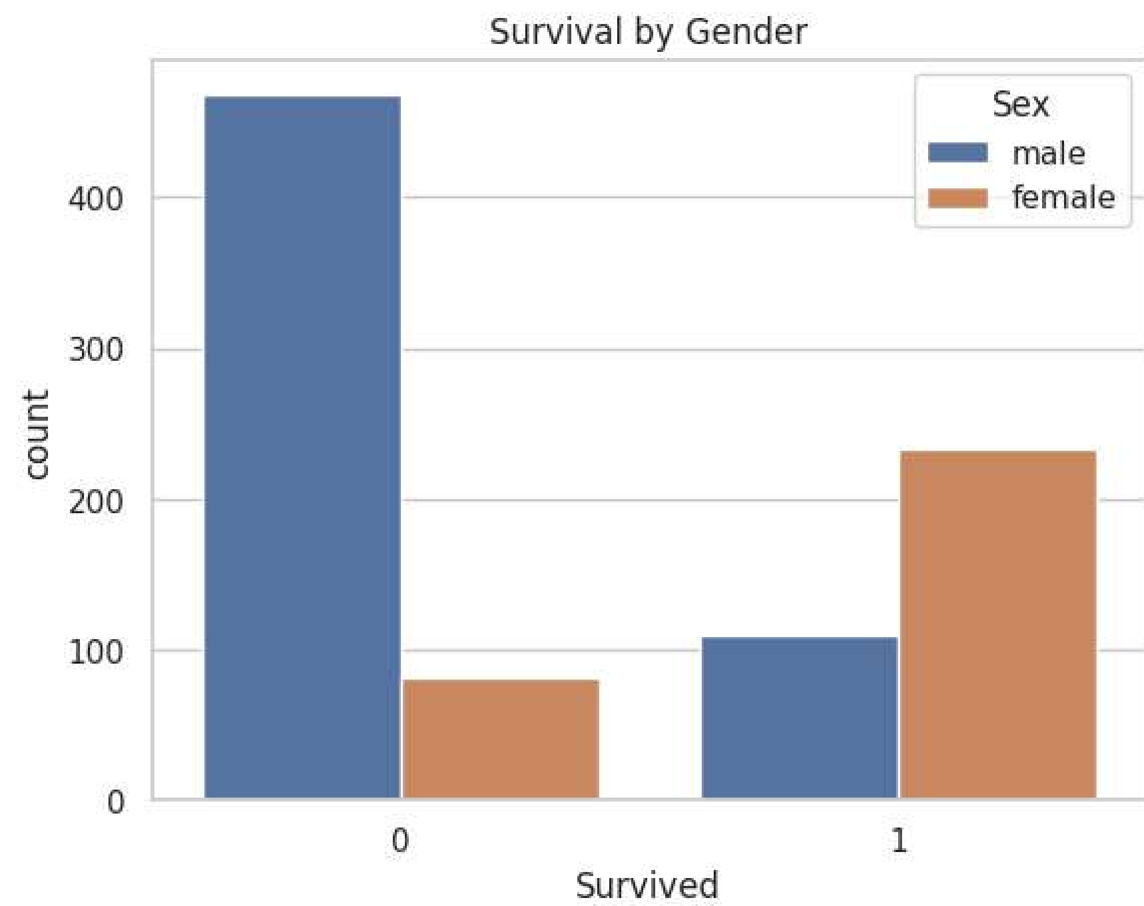


Observation:

Fare distribution contains several extreme outliers.

Data is highly right-skewed.

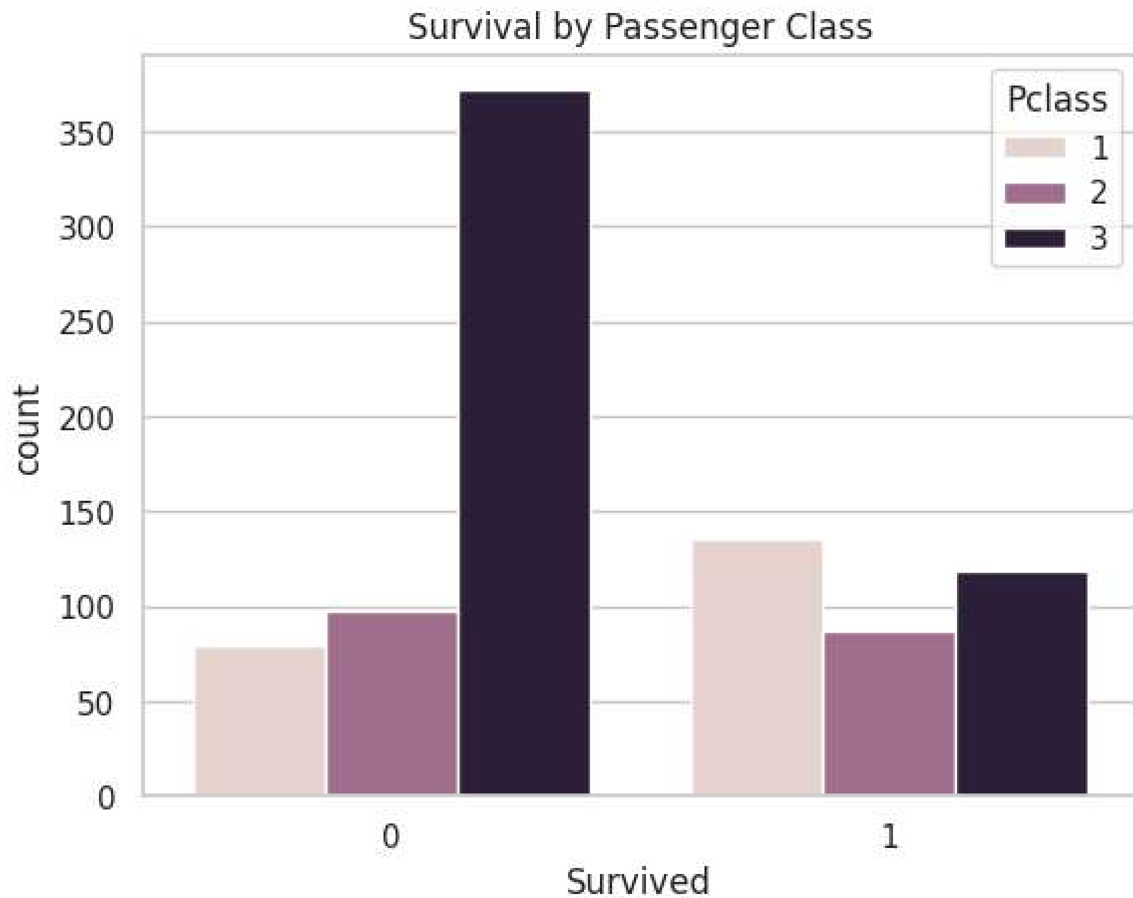
```
sns.countplot(x='Survived', hue='Sex', data=df)
plt.title("Survival by Gender")
plt.show()
```



Observation:

Female passengers had a significantly higher survival rate than males.

```
sns.countplot(x='Survived', hue='Pclass', data=df)
plt.title("Survival by Passenger Class")
plt.show()
```

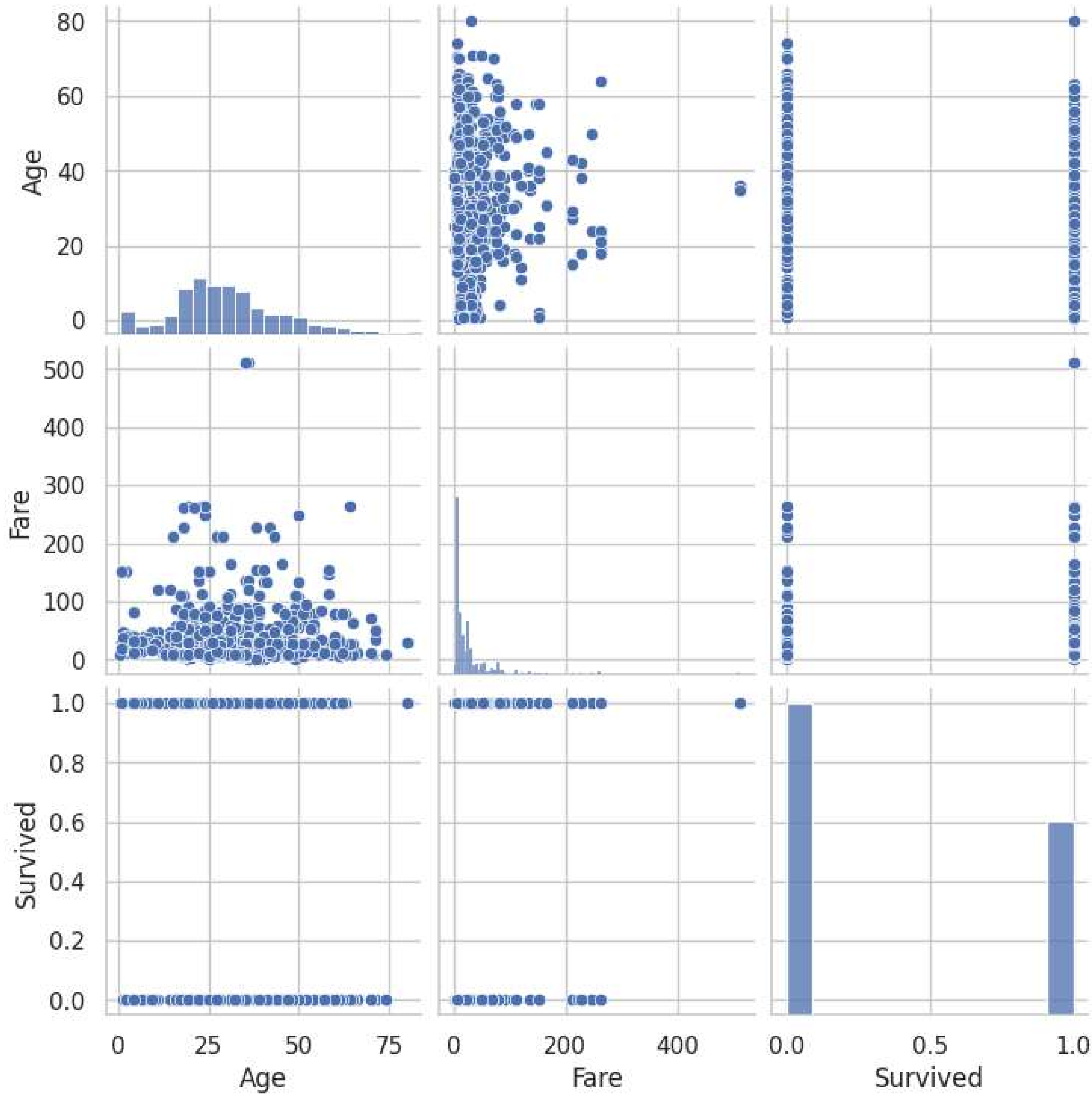


Observation:

First-class passengers had the highest survival rate.

Survival probability decreases as passenger class decreases.

```
sns.pairplot(df[['Age', 'Fare', 'Survived']])
plt.show()
```



Observation:

Higher fare values are associated with higher survival.

No strong linear relationship observed with age.

```
plt.figure(figsize=(8,6))
sns.heatmap(df.select_dtypes(include=np.number).corr(), annot=True, cmap='coolw
plt.title("Correlation Heatmap")
plt.show()
```